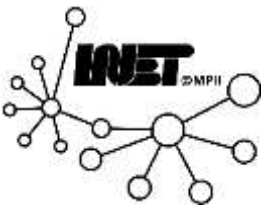




# Exploring the Cookieverse: A Multi-Perspective Analysis of Web Cookies

Devashish Gosain



**KU LEUVEN**

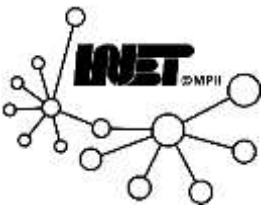


## A multi perspective analysis of Web Cookies (IMC 23, PAM 23)

# Cookies---Threat to Privacy?



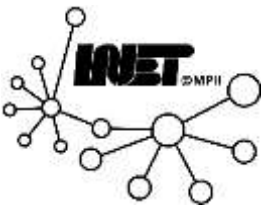
The screenshot displays the Spotify web interface. On the left is a dark sidebar with navigation links: Home, Search, Your Library, Create Playlist, and Liked Songs. Below these are links for Legal, Privacy Center, Privacy Policy, Cookie Settings, About Ads, and Cookies, along with a language selector set to English. The main content area is titled 'Focus' and features a grid of seven playlist cards: 'Peaceful Piano', 'Deep Focus', 'Instrumental Study', 'Jazz Vibes', 'Focus Flow', 'Workday Lounge', and 'Beats to think to'. Each card includes a cover image and a brief description. Below the 'Focus' section is a 'Spotify Playlists' section with another grid of cards: 'TODAY'S TOP', 'RapCaviar', 'All Out 2010s', 'Rock Classics', 'Chill Hits', 'VIVA LATINO', and 'Mega Hit Mix'. At the bottom, a purple banner reads 'PREVIEW OF SPOTIFY Sign up to get unlimited songs and podcasts with occasional ads. No credit card needed.' and a blue button says 'Sign up free'.



# Cookies---Threat to Privacy?



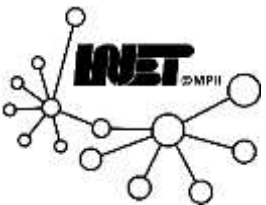
The screenshot shows the top of the Washington Post website. The navigation bar is black with white text for categories: Politics, Opinions, War in Ukraine, Investigations, and Climate. On the right of the bar are 'Subscribe' and 'Sign in' buttons. Below the bar, the site's masthead features 'The Washington Post' in a large serif font, with the tagline 'Democracy Dies in Darkness' underneath. The main content area displays two advertisements. The first is a yellow ad for 'SPARKS MUSIC' featuring a pair of blue headphones and a '30% OFF' discount. The second is a blue ad with the text 'Go Premium. Be happy.' and a photo of a smiling woman.



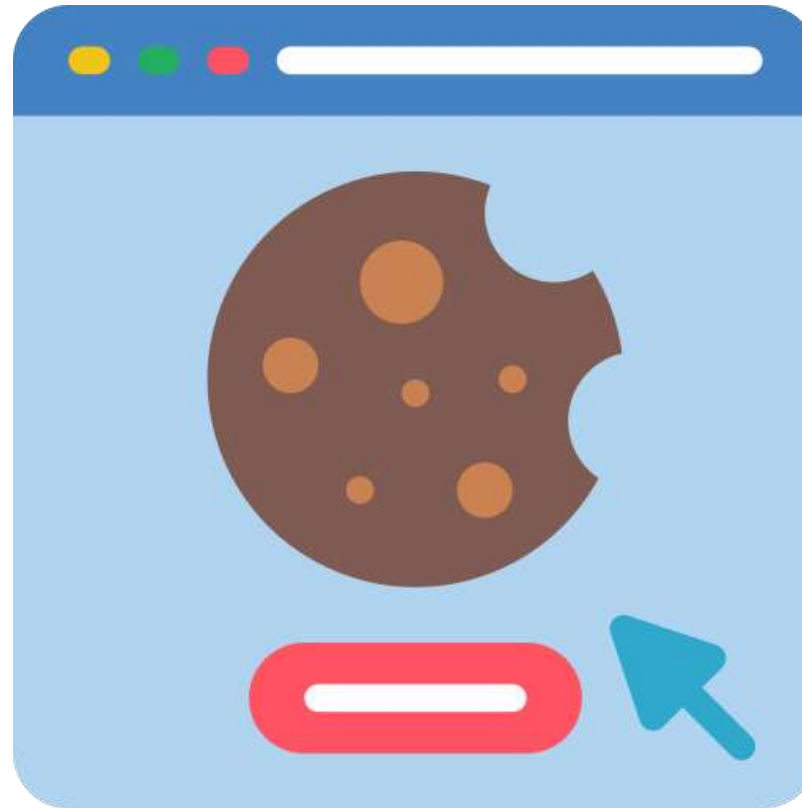
# Cookies---Threat to Privacy?



The screenshot shows a news website with a dark navigation bar at the top. The bar contains a search icon, a bell icon, and several category links: Politics, Opinions, War in Ukraine, Investigations, and Climate. On the right side of the bar are 'Subscribe' and 'Sign in' buttons. Below the navigation bar, there is a yellow banner featuring a pair of blue headphones and the text 'Lorem ipsum dolor consectetur adipiscing elit, sed diam.' Below this is a large advertisement for Spotify Premium, which includes the text 'Go Premium. Be happy.' and 'Premium sounds amazing. Enjoy ad-free music on all your devices.\*' with a 'GET SPOTIFY PREMIUM' button. At the bottom of the page, there is a news article titled 'Washington's bank rescue fails to erase all doubts'. To the right of the article title is a section titled 'Bank failures since 2007' which lists 'Washington Mutual' with a value of '\$307B'. The name 'Signature Bank' is visible at the bottom right of the page.



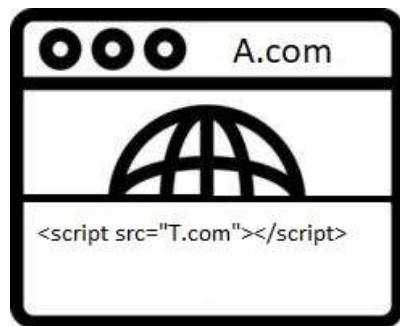
# Cookies---Threat to Privacy?





**T.com**  
(jh234)

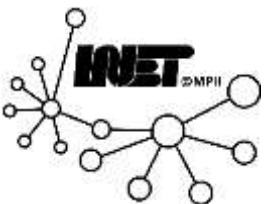
**A.Com**  
(lkcv4)



**T.com**



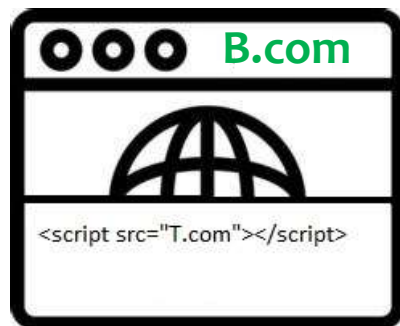
**A.com**



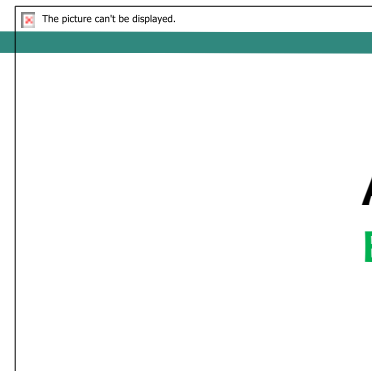


T.com  
(jh234)

B.Com  
(fg56a)

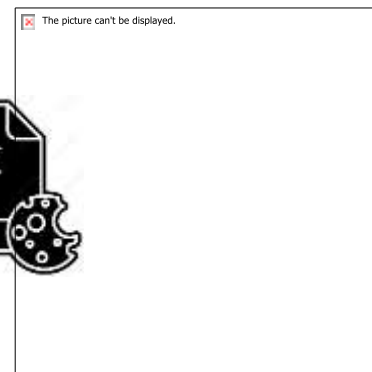


jh234

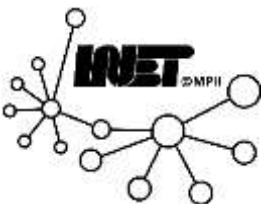


T.com

A.com  
B.com



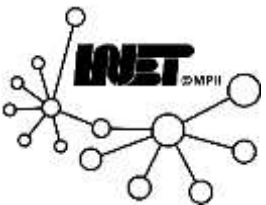
B.com



# Privacy Laws

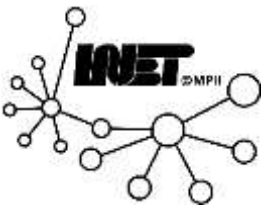
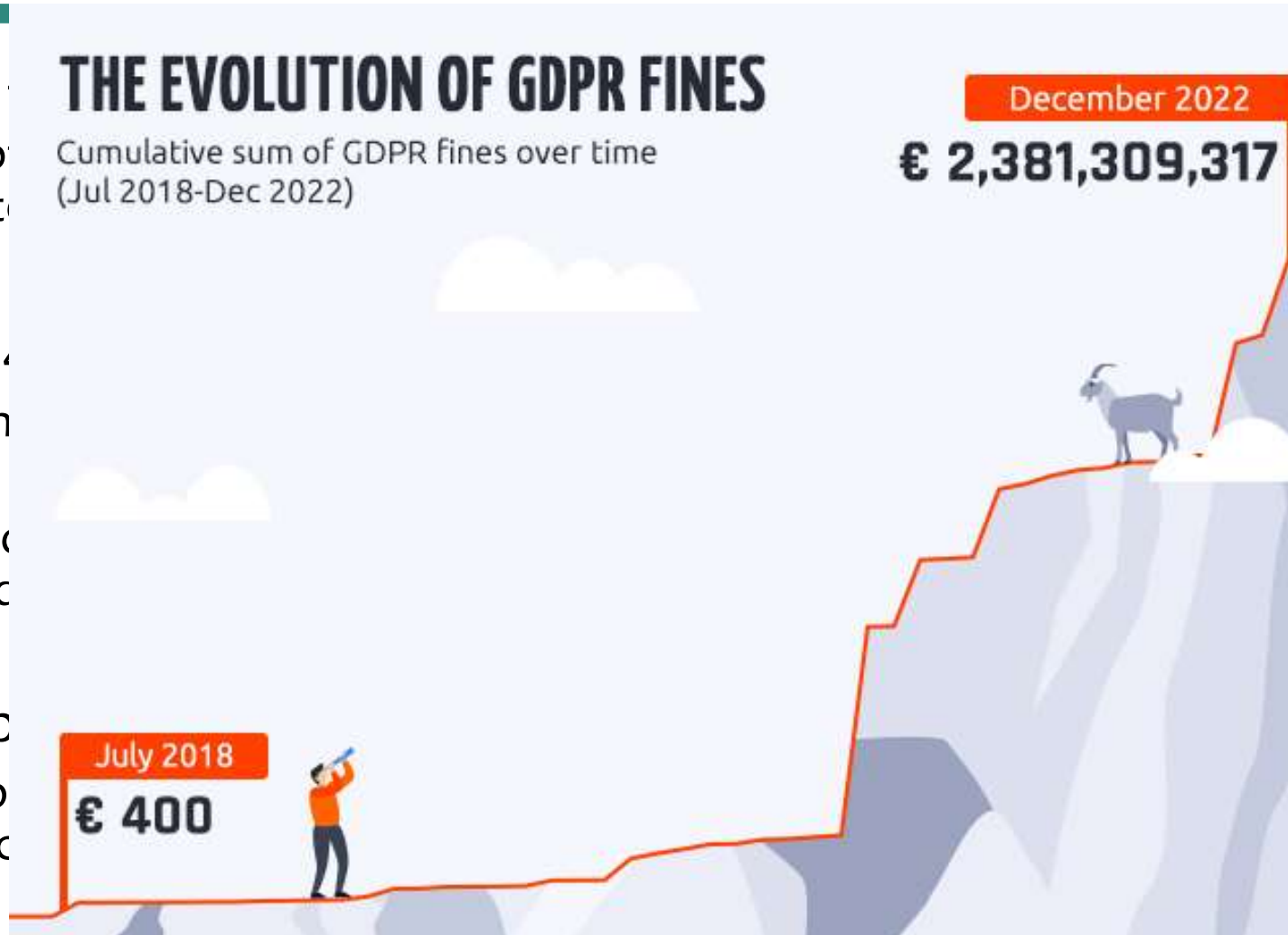


- General Data Protection Regulation (GDPR)
  - EU law
  - May 25, 2018
  - Enhance individuals' control and rights over their personal data
- California Consumer Privacy Act (CCPA)
  - California State law
  - January 1, 2020



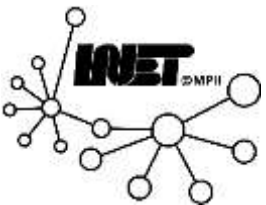
# GDPR violations (examples)

- Amazon
  - It was not before started
- Meta – €4.55 billion
  - Instagram 13 and 17
  - It violated email add
- WhatsApp
  - WhatsApp data proc notice





Website publishers use **Cookie Banners**



# Cookie Banners (notices)



The screenshot shows the Spotify website interface. At the top, there's a navigation bar with the Spotify logo, navigation links (Home, Search, Your Library, Create Playlist, Liked Songs), and user options (Premium, Support, Download, Sign up, Log in). Below the navigation bar, there's a 'Focus' section with several music playlists: 'Peaceful Piano', 'Deep Focus', 'Instrumental Study', 'Jazz Vibes', 'Focus Flow', 'Workday Lounge', and 'Beats to think to'. A blue-bordered cookie banner is overlaid on the bottom half of the page. The banner contains text in German explaining the use of cookies and providing options to accept or manage cookies.

**Deine Privatsphäre ist uns wichtig**

Wir und unsere Partner nutzen Cookies, um personenbezogene Daten wie z.B. Browsing-Daten zu speichern und abzurufen, um z.B. Inhalte und Werbung bereitzustellen und zu personalisieren sowie die Verwendung der Website zu analysieren. Du erfährst mehr über die Zwecke, für welche wir und unsere Partner Cookies einsetzen, und du kannst deine Einstellungen ändern, wenn du unten auf den Button „Cookie Einstellungen“ klickst. Du kannst jederzeit deine Cookie-Auswahl überdenken oder deine Einwilligung widerrufen, indem du auf den Link zu den Cookie-Einstellungen in unserer Cookie-Policy klickst. Wir arbeiten auf Grundlage eines Industriestandards, wodurch wir deine Einstellungen weltweit an alle teilnehmenden Websites weitergeben.

Indem Du auf „Cookies akzeptieren“ klickst, willigst Du in unsere Nutzung und die Weitergabe Deiner Daten an unsere Partner ein.

**Wir und unsere Partner verarbeiten Daten, um Folgendes bereitzustellen:**

Informationen auf einem Gerät speichern und/oder abrufen. Personalisierte Anzeigen. Personalisierte Inhalte. Anzeigenmessung, Inhaltmessung, Erkenntnisse über Zielgruppen und Produktentwicklung.

Weitere Informationen zu unseren Partnern und zum Opt-out findest Du in unserer:

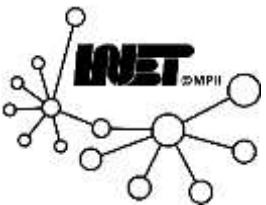
[LISTE DER PARTNER \(ANBIETER\)](#)

**COOKIES AKZEPTIEREN** **COOKIE EINSTELLUNGEN**

# Research Questions



- How many websites do show the banners?
  - Does the banner give users the options to explicitly agree or decline?
- Do they respect the users preferences?
- Do websites exhibit different behavior?
  - Geographic location (EU vs. non-EU)
  - User agent (mobile vs. desktop)



# Related Works

- Not considering users' characteristics
- Manual or semi-automated inspection
- Hard-to-scale automated tool
- Just “accept” option

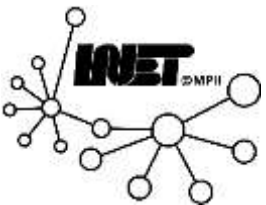


# Goals



## Analysis of cookie landscape from different perspectives in a automated way

- Detection of cookie banners
- Interaction with cookie banners
- Impact of geographic locations
- Consistency of websites
- Cookie differences between
  - landing and inner pages
  - Mobile and desktop
- Impact of CCPA

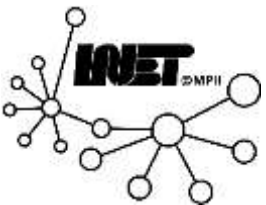


# Goals



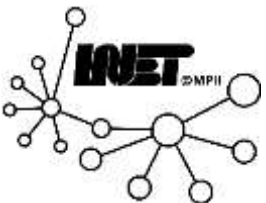
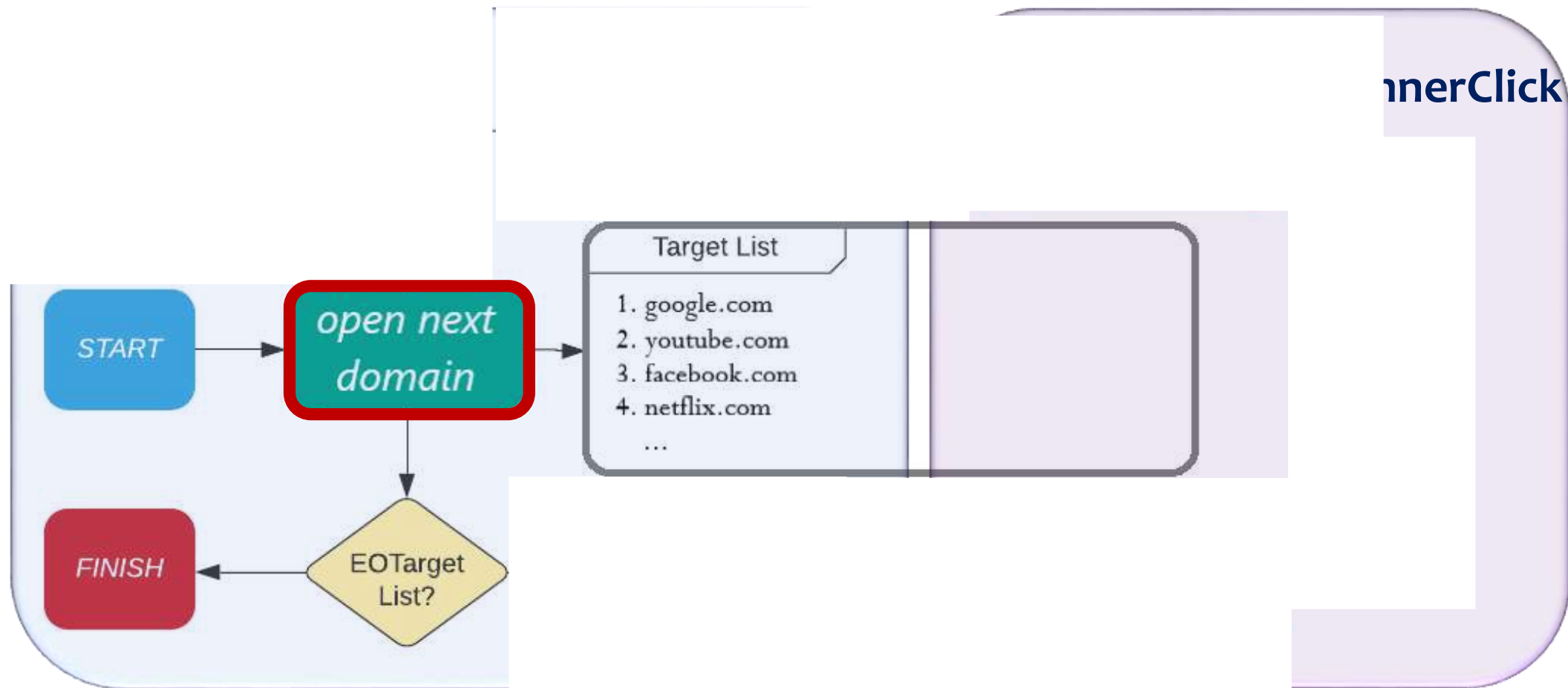
## Analysis of cookie landscape from different perspectives in a automated way

- Detection of cookie banners
- Interaction with cookie banners
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  - Mobile and desktop
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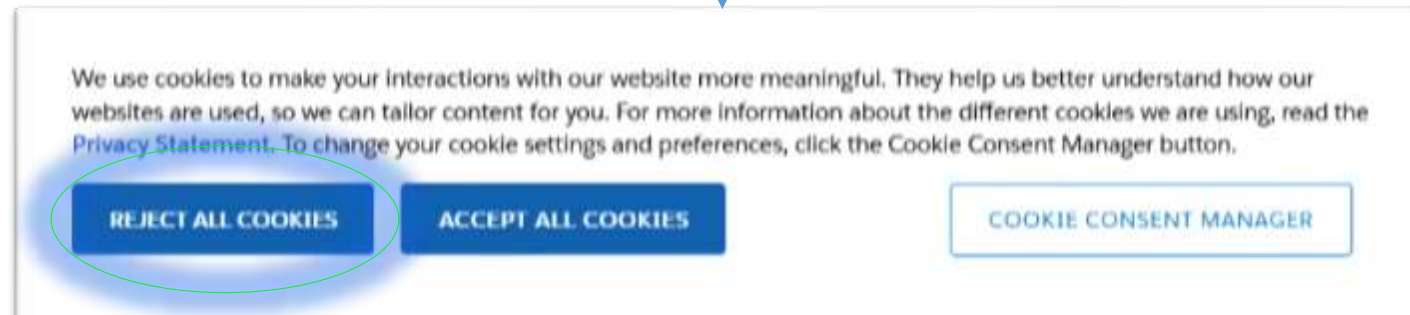
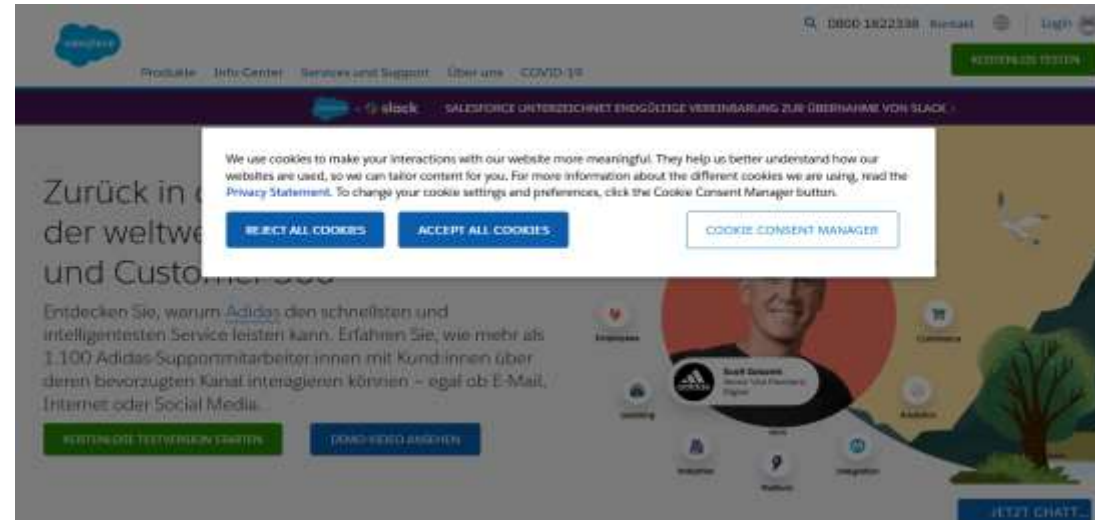


# Methodology





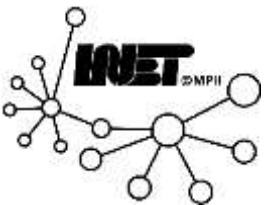
# BannerClick



# Research Challenges

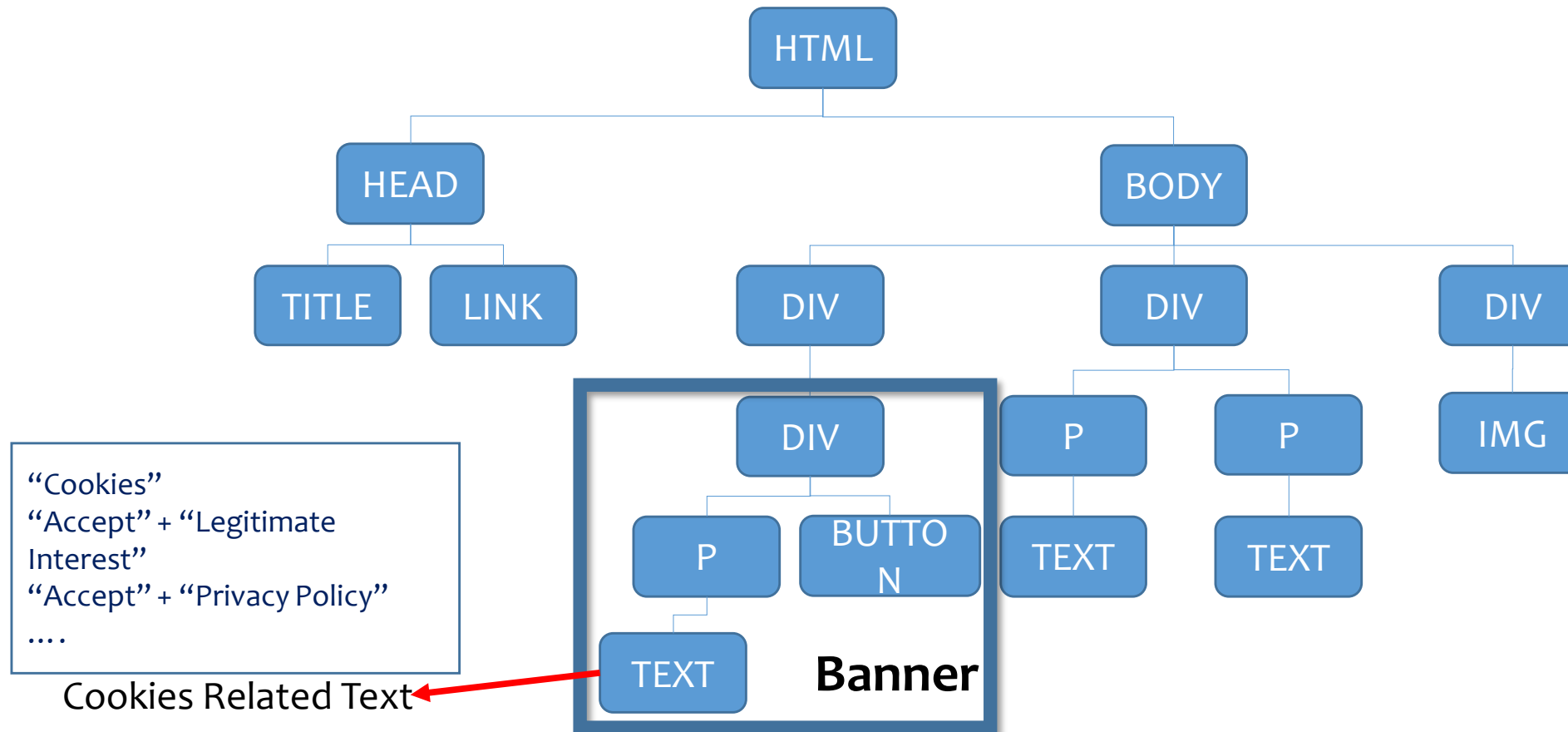


- There is no standardization of banners; they differ in:
  - Shape & size
  - Location (top, bottom, middle etc.)
  - Content
  - Language
- Websites with Consent Management Platform (CMPs)
  - CMP are dedicated contractors just to manage banners
  - They use different APIs to show banner
- Banners adhere to different privacy laws differently
  - Banners respecting GDPR show “accept” and “reject” options
  - Banners respecting CCPA contains “DNSMPI” link





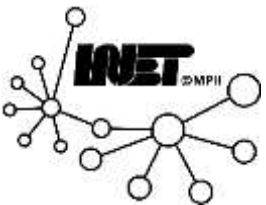
# BannerClick – Detection



# Cookie Related Words



- Created a corpus of English words by inspecting 50 random websites from Tranco top 100 domains
  - Total 8 English words: cookies, privacy, policy, consent, accept, agree, personalized, and legitimate interest
- Translates these words into 11 different languages
  - German, Swedish, Spanish, Italian, Portuguese, Chinese, Russian, Japanese, French, Turkish, and Persian
  - Corpus size: 80 words

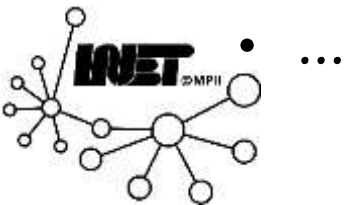


# Are all elements with cookie related words part of banner?



If an element has words from our corpus **BUT** has the following property:

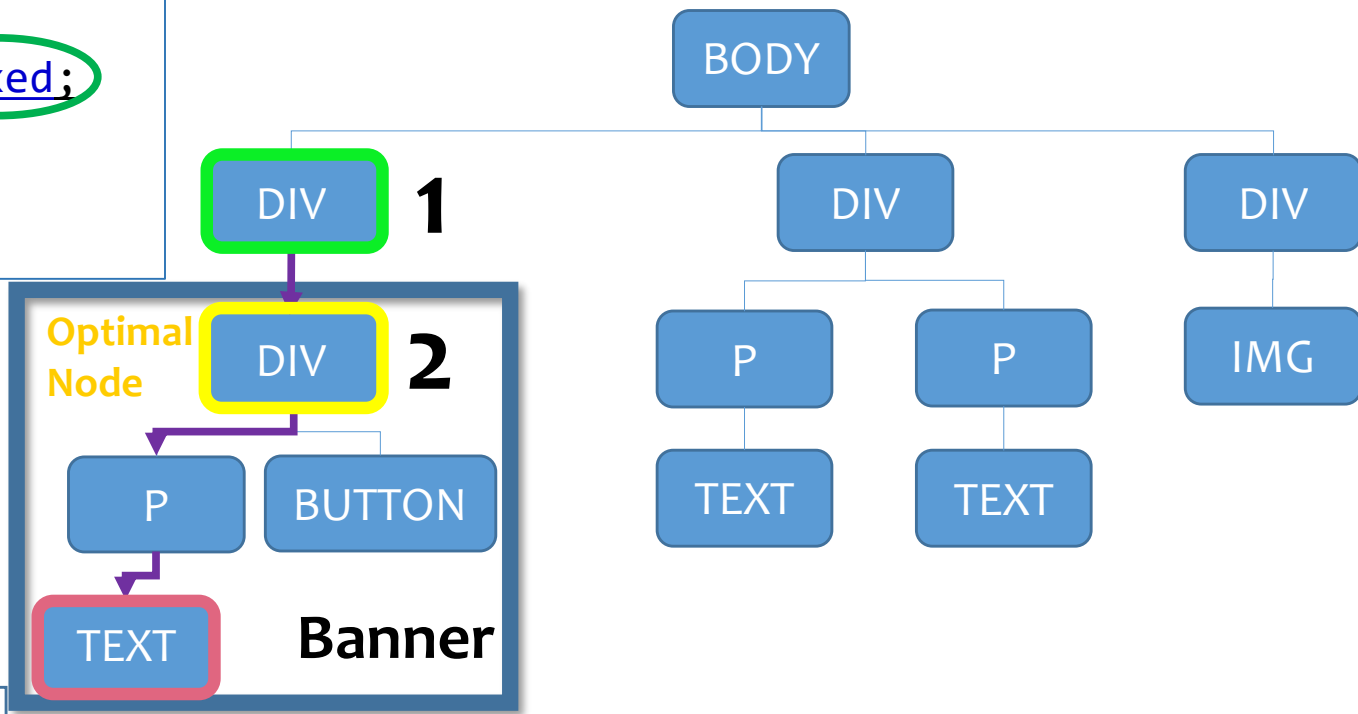
- Element is set as invisible
  - the banner is not visible to users, and they can therefore not interact with it
- Element with a negative z-index
  - It is behind some other objects on the page
  - It cannot contain a banner; banner should be on top of every object
- Element outside the viewport cannot contain the banner
  - The banner should be within the user's visible area of a web page
- ...





# BannerClick – Detection

```
div.firstnode {  
  position: fixed;  
  width: 300px;  
  border: 3px  
}
```



End Node

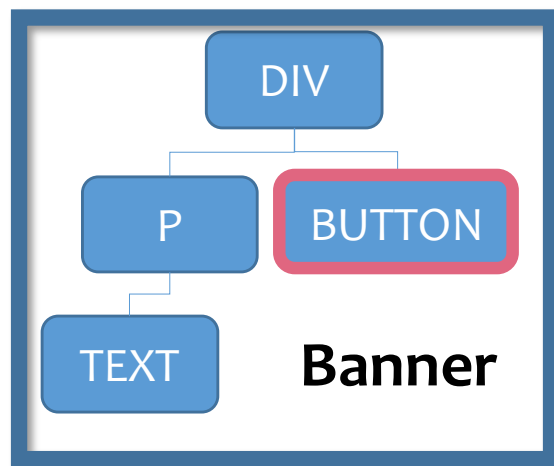
- “Cookies”
- “Accept” + “Legitimate Interest”
- “Accept” + “Privacy Policy”
- ....

99%  
Accuracy





# BannerClick – Interaction



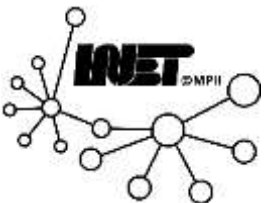
97%  
Accuracy

Word sets:

acc: ["accept", "agree", "confirm"]

rej: ["reject", "refuse", "decline"]

set: ["setting", "option", "choice"]

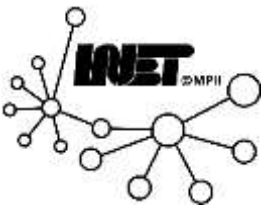




# Measurement Setup

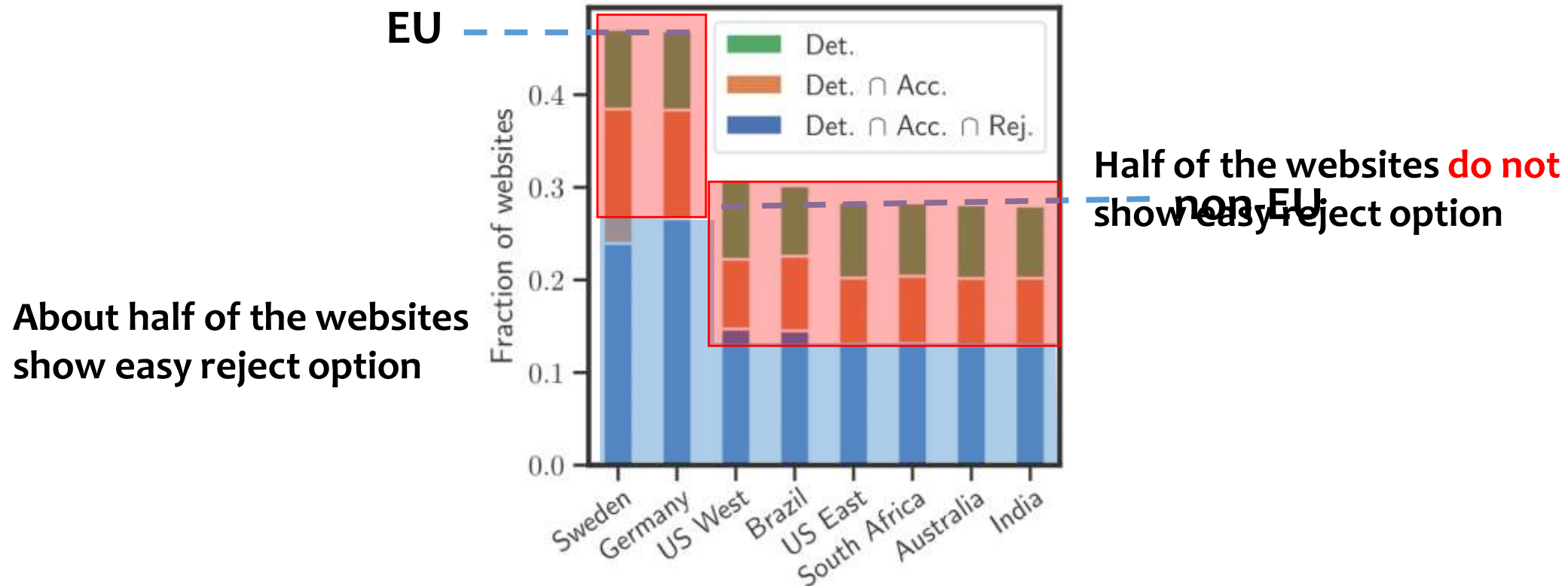


- **8 vantage points:** Germany, Sweden, US West, US East, India, Brazil, South Africa, Australia
- **Target list:** Tranco Top 10k domains





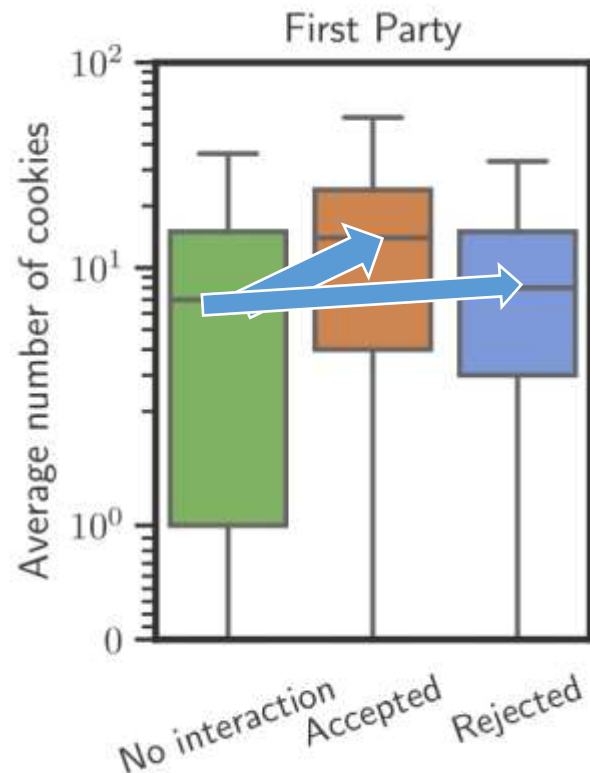
# Banners Detected, Accepted, Rejected



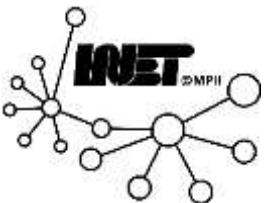
More banners in EU compared to non-EU countries



# Cookies Differences After Interaction

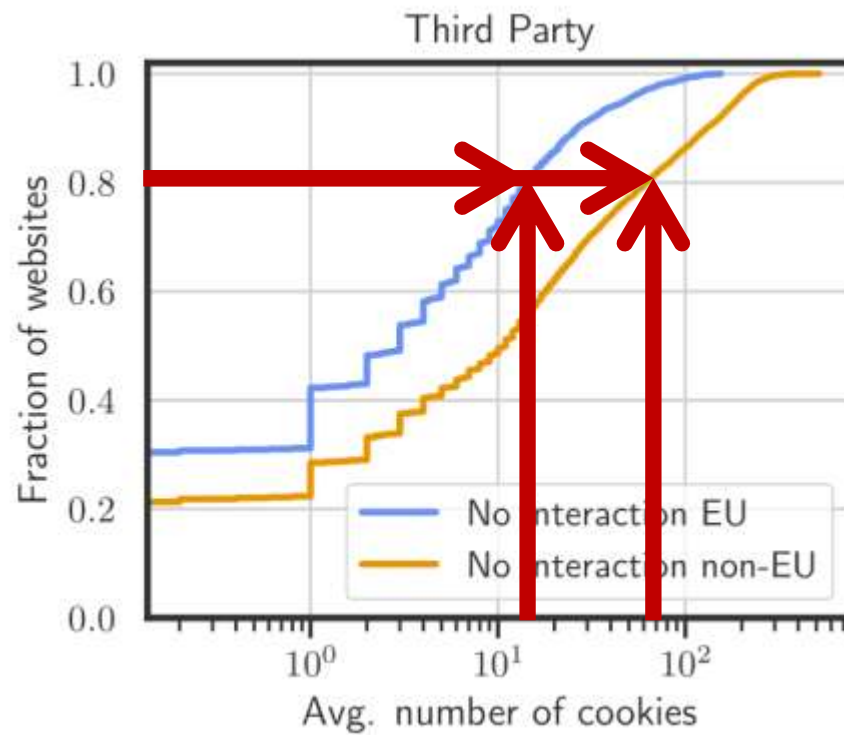


**Interacting with banners impacts cookie distribution**





# Cookies – EU vs. non-EU

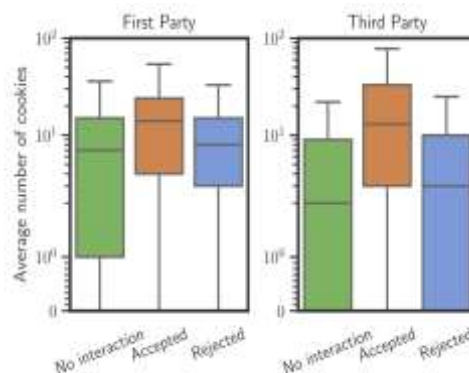
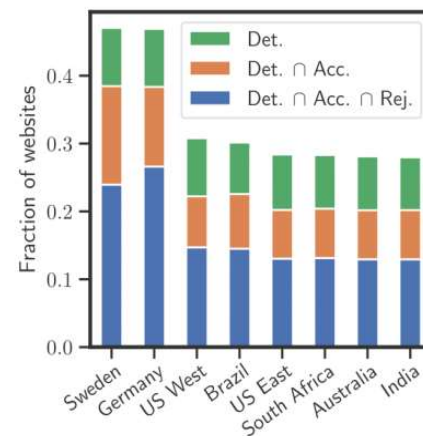


**Fewer cookies in EU compared to non-EU**

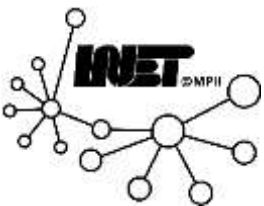
# Conclusion



- Impact of
  - Geographical location of users
  - Interaction with banner
- Check out the paper
  - CCPA impact
  - Landing vs. Inner pages
  - Mobile vs. Desktop
  - Consistency analysis
- Source code available
  - BannerClick
  - Analysis data for reproducibility

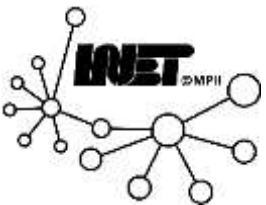


[bannerclick.github.io](https://bannerclick.github.io)





## Part II: Emergence of Accept-Or-Pay Cookie Banners



# The Rise of Accept-Or-Pay Cookie Banners



Montag, April 24, 2023 Datenschutzerklärung | Impressum [f](#) [in](#) [t](#) [v](#)

## Bahnblogstelle

Nachrichten aus der Welt der Eisenbahn

24. April 2023, 04:17

### Frau verhindert Zugabfahrt und Zugreisender greift ein

**B**

#### Consent to Cookies & Data processing

Visit this website as usual with advertising and tracking. You can revoke your consent at any time via the privacy policy.

☒ Accept all

More Info on advertising and tracking can be found in our privacy policy or in our privacy center.

... or with contentpass

Visit this website completely free of advertising, personalized tracking and commercials from 2.99 € / month.

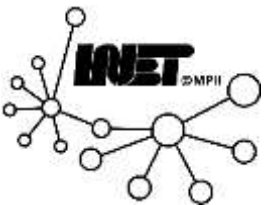
☒ Ad-free for 2.99 € / month

Already contentpass member? ☒ Login

On this website we use cookies and similar functions to process end device information and personal data. The processing is used for purposes such as to integrate content, external services and elements from third parties, statistical analysis/measurement, personalized advertising and the integration of social media. Depending on the function, data is passed on to up to 3 third parties and processed by them. This consent is voluntary, not required for the use of our website and can be revoked at any time using the icon on the bottom left.

☒ Store and/or access information on a device powered by

36-jähriger fährt auf Waggon von Autozug mit - Bahnstrecke gesperrt



# Cookiewalls



## Consent to Cookies & Data processing

Visit this website as usual with advertising and tracking. You can revoke your consent at any time via the privacy policy.

✓ Accept all

More Info on advertising and tracking can be found in our privacy policy or in our privacy center.

3

## ... or with contentpass

Visit this website completely free of advertising, personalized tracking and commercials from 2.99 € / month.

Ad-free for 2.99 € / month

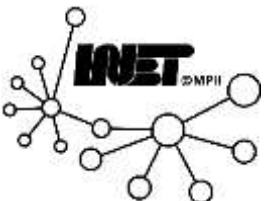
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On this website we use cookies and similar functions to process end device information and personal data. The processing is used for purposes such as to integrate content, external services and elements from third parties, statistical analysis/measurement, personalized advertising and the integration of social media. Depending on the function, data is passed on to up to 3 third parties and processed by them. This consent is voluntary, not required for the use of our website and can be revoked at any time using the icon on the bottom left.

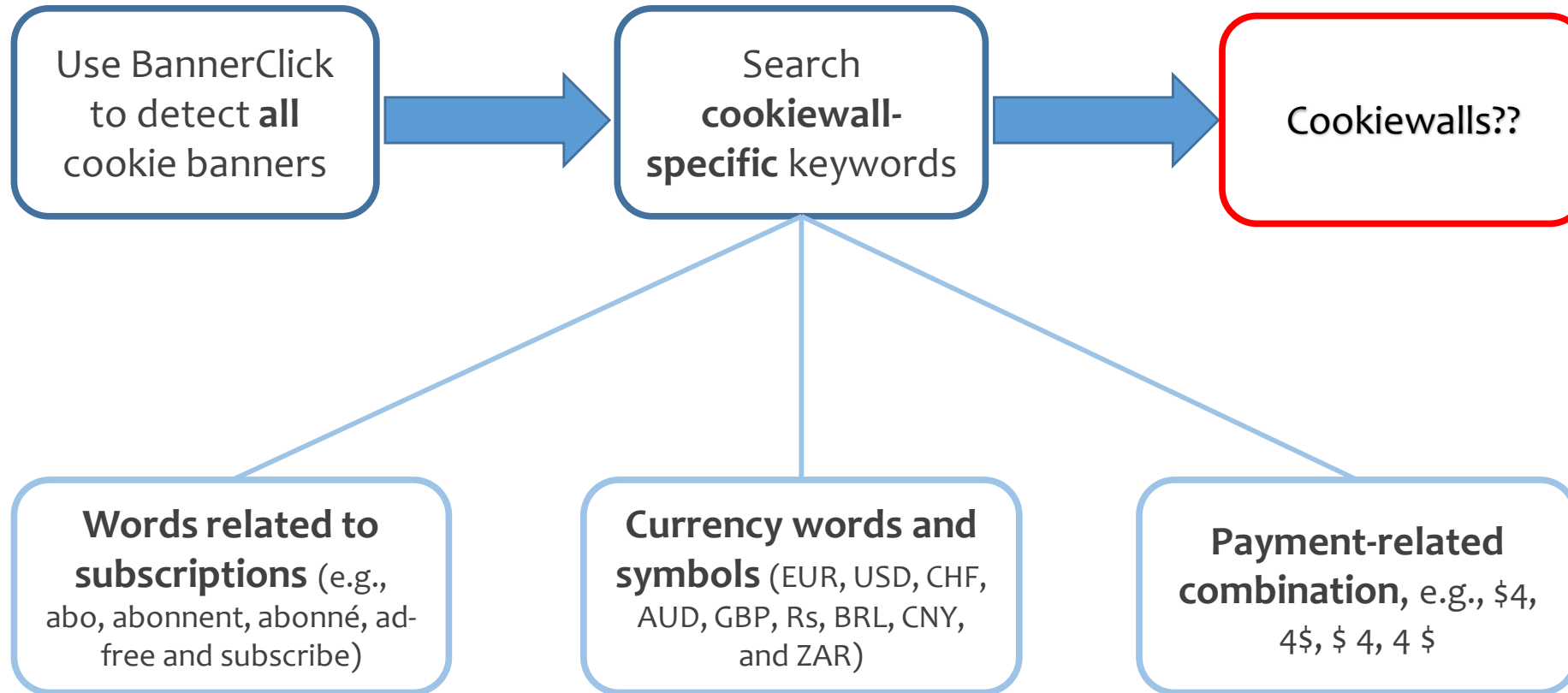


Store and/or access information on a device

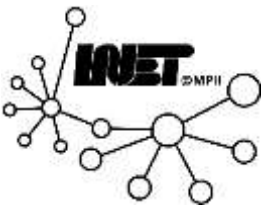
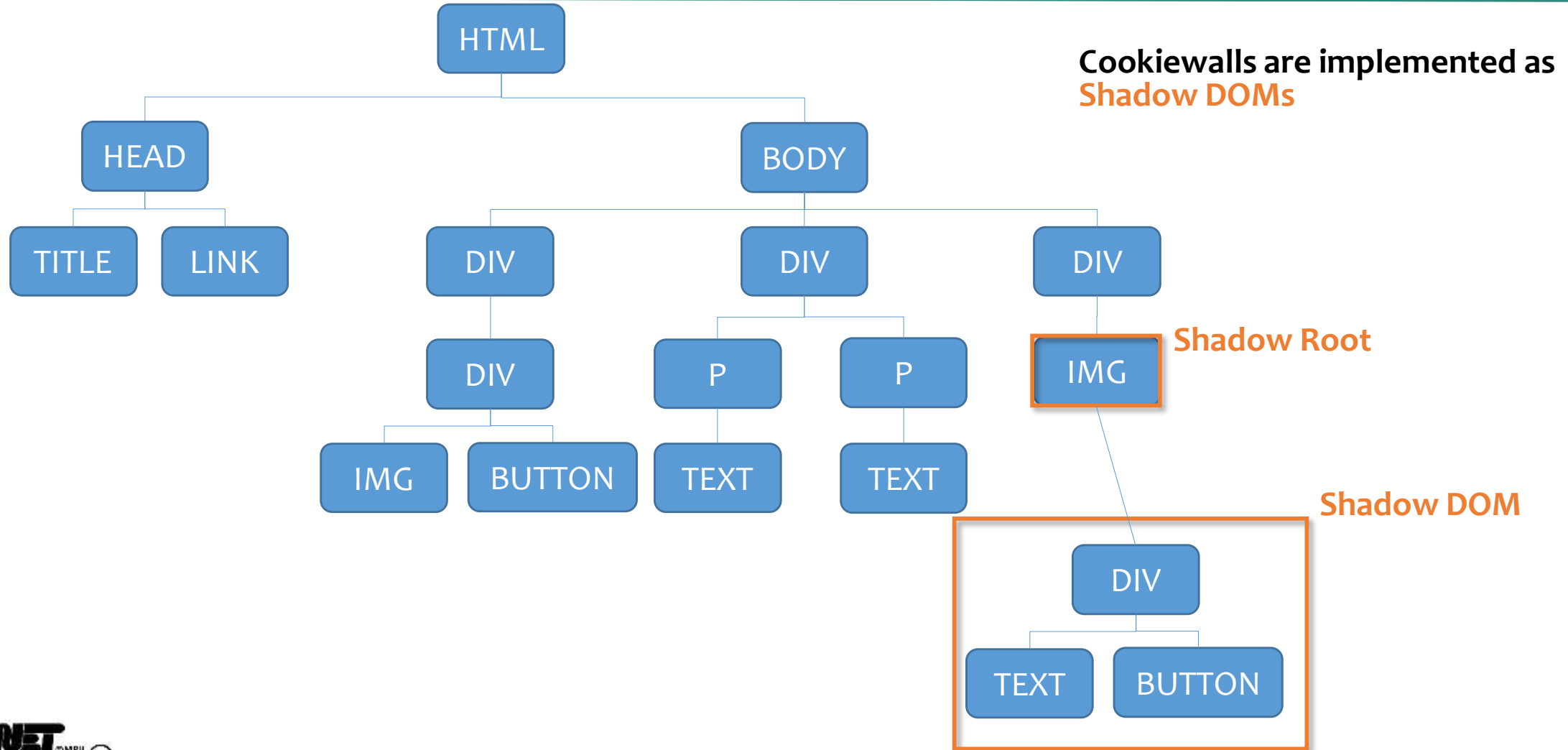
powered by



# Methodology



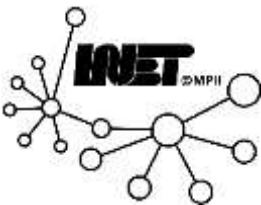
# What We Missed?



# BannerClick++



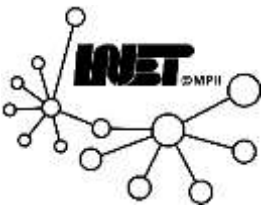
- In the HTML DOM search for elements with the **shadow\_root** property
- Clone and append all child elements within a shadow DOM to the body element of the main document DOM
- Then we use the original BannerClick module
- Accuracy:
  - For 1k randomly selected domains manually checked their screenshots to find cookiewalls on the website
  - Also, we manually checked all 285 websites where we detected a cookiewall
  - Precision and Recall: >99%



# Measurement Setup



- Used Google's Chrome User Experience Report (CrUX) for target selection
  - Better toplist compared to Alexa or Tranco
- Take union of the country-wise Google CrUX top 10k domains for each vantage point (VP)
  - **45,222** unique domains reachable in all VPs

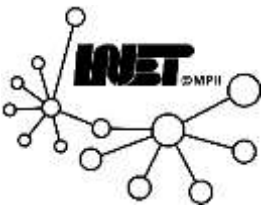


# Cookiewalls Detected

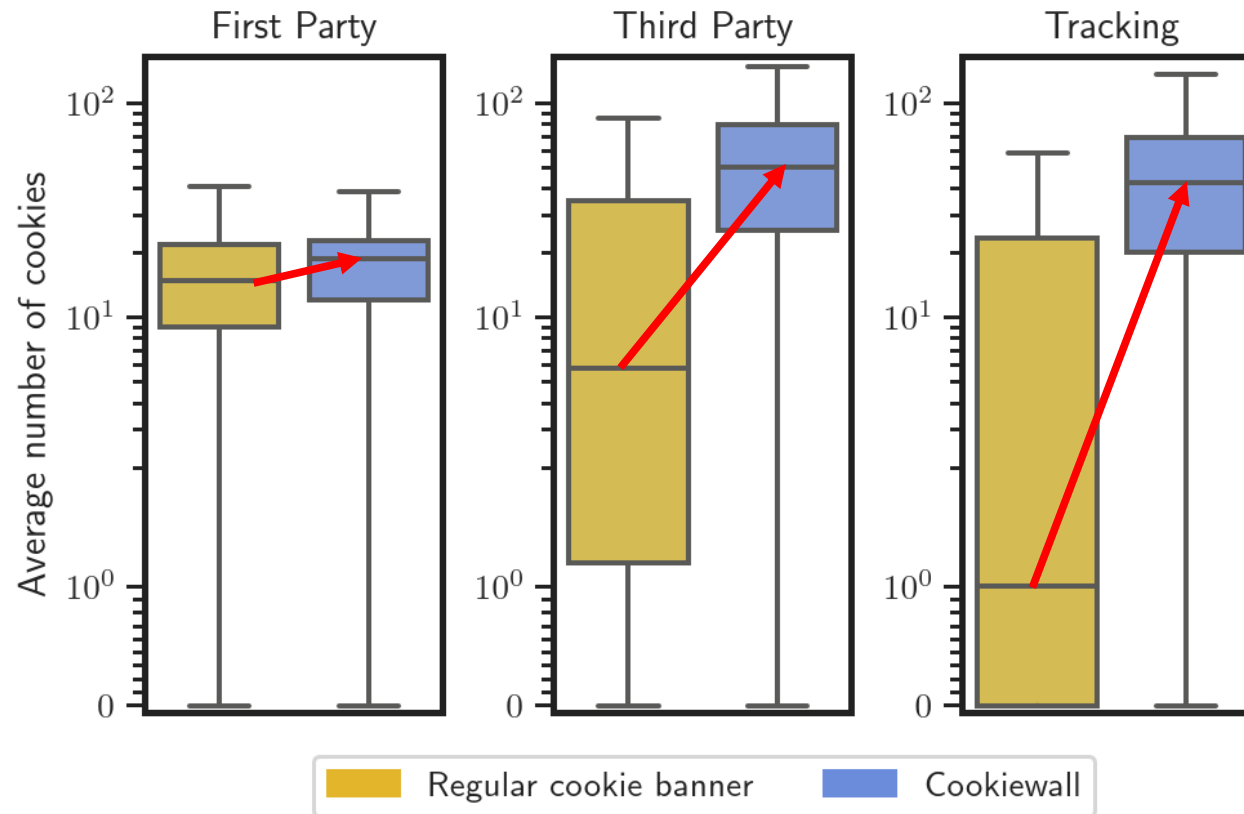


| VP           | Cookiewalls | Toplist | ccTLD | Language |
|--------------|-------------|---------|-------|----------|
| US East      | 197         | 0       | 0     | 9        |
| US West      | 199         | 0       | 0     | 9        |
| Brazil       | 196         | 0       | 0     | 0        |
| Germany      | 280         | 259     | 233   | 252      |
| Sweden       | 276         | 15      | 0     | 0        |
| South Africa | 199         | 0       | 0     | 0        |
| India        | 192         | 0       | 0     | 10       |
| Australia    | 190         | 5       | 0     | 10       |

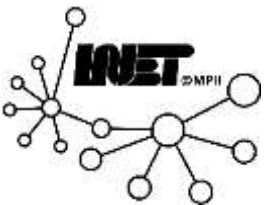
**Table 1: Number of detected cookiewalls depending on the country of the vantage point, country-specific toplist, TLD associated with that country, and the most commonly spoken language in that country.**



# Cookie Difference: Regular Banner vs Cookiewall



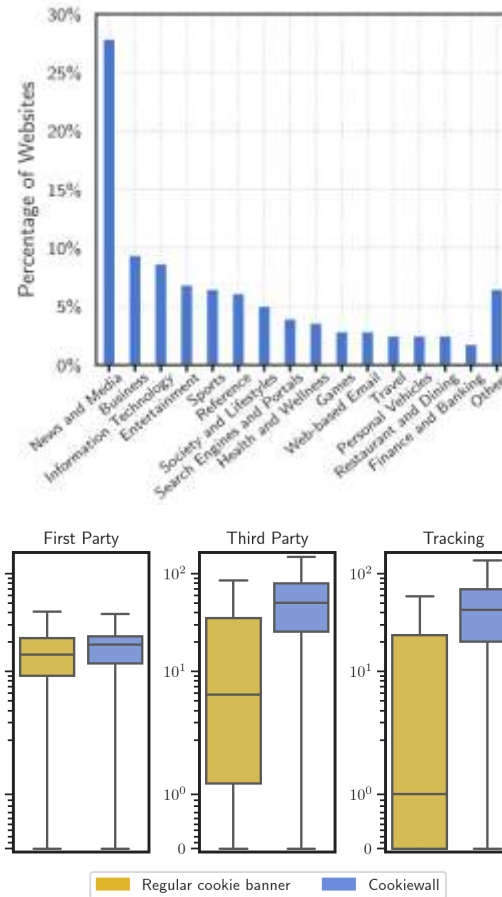
**Cookiewalls set more third-party and tracking cookies than regular banners**



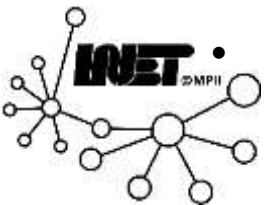
# Conclusion



- Analysis of Cookiewalls
  - Automatically detect cookiewalls with a precision of >99%
  - Crawled 45k websites; found cookiewalls on 280 of them
  - News and Media websites show max. cookiewalls
  - Cookiewalls send 42 times more tracking cookies than regular cookie banners
- Check out the paper
  - Subscription Management Platforms
- Source code available
  - Data is publicly available



[bannerclick.github.io](https://bannerclick.github.io)





# Backup: Measurement Setup

- Main run
  - Tranco Top 10k List domains → 9020 totally reachable
  - 5 iteration for each mode of interaction
- Desktop vs. mobile (and others)
  - Tranco tiered 300 → top-100, 1001–1100, and 9901–10k
  - User Agent
    - Mobile: “Mozilla/5.0 (Android 12; Mobile; rv:68.0) Gecko/68.0 Firefox/93.0”
    - Desktop: “Mozilla/5.0 (X11; Linux x86\_64; rv:95.0) Gecko/20100101 Firefox/95.0”
  - Screen size
    - Mobile: 340x695
    - Desktop: 1366x768





# Backup: Measurements

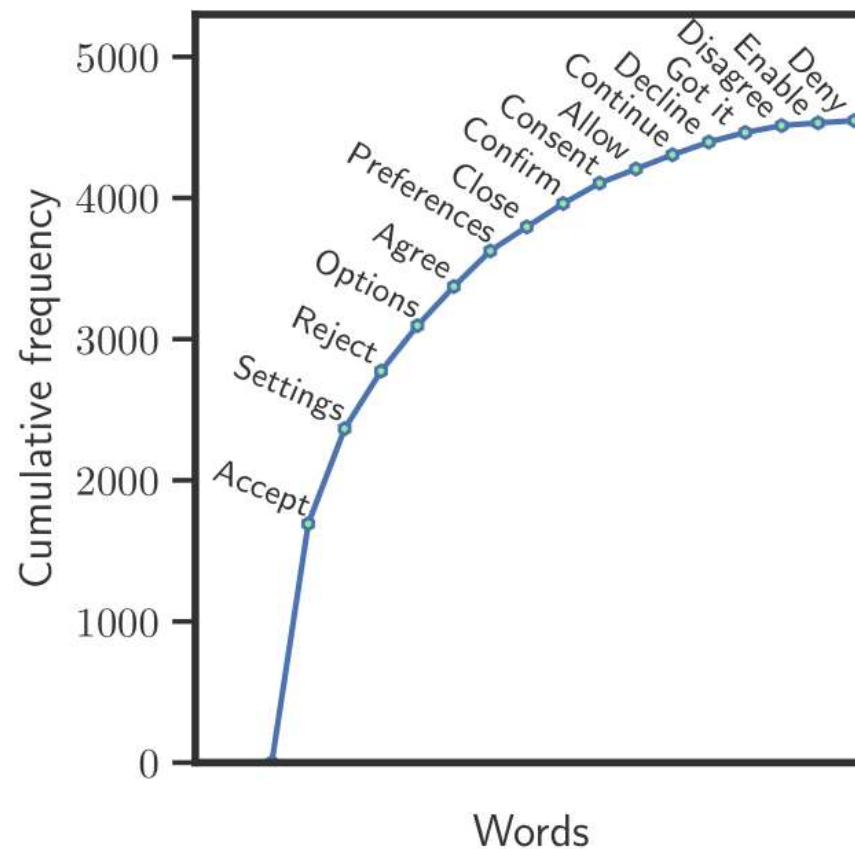
**Table 1.** Overview of different measurement types.

| Measurement Type   | Start Date   | Duration | Target Websites   |
|--------------------|--------------|----------|-------------------|
| Banner Interaction | Jan 20, 2022 | 20 days  | Tranco Top 10k    |
| Consistency Tests  | Feb 9, 2022  | 10 days  | Tranco tiered 300 |
| Landing vs. Inner  | Mar 8, 2022  | 4 days   | Tranco tiered 300 |
| Desktop vs. Mobile | Feb 27, 2022 | 10 hours | Tranco tiered 300 |
| Impact of CCPA     | Mar 13, 2022 | 10 hours | Tranco tiered 300 |



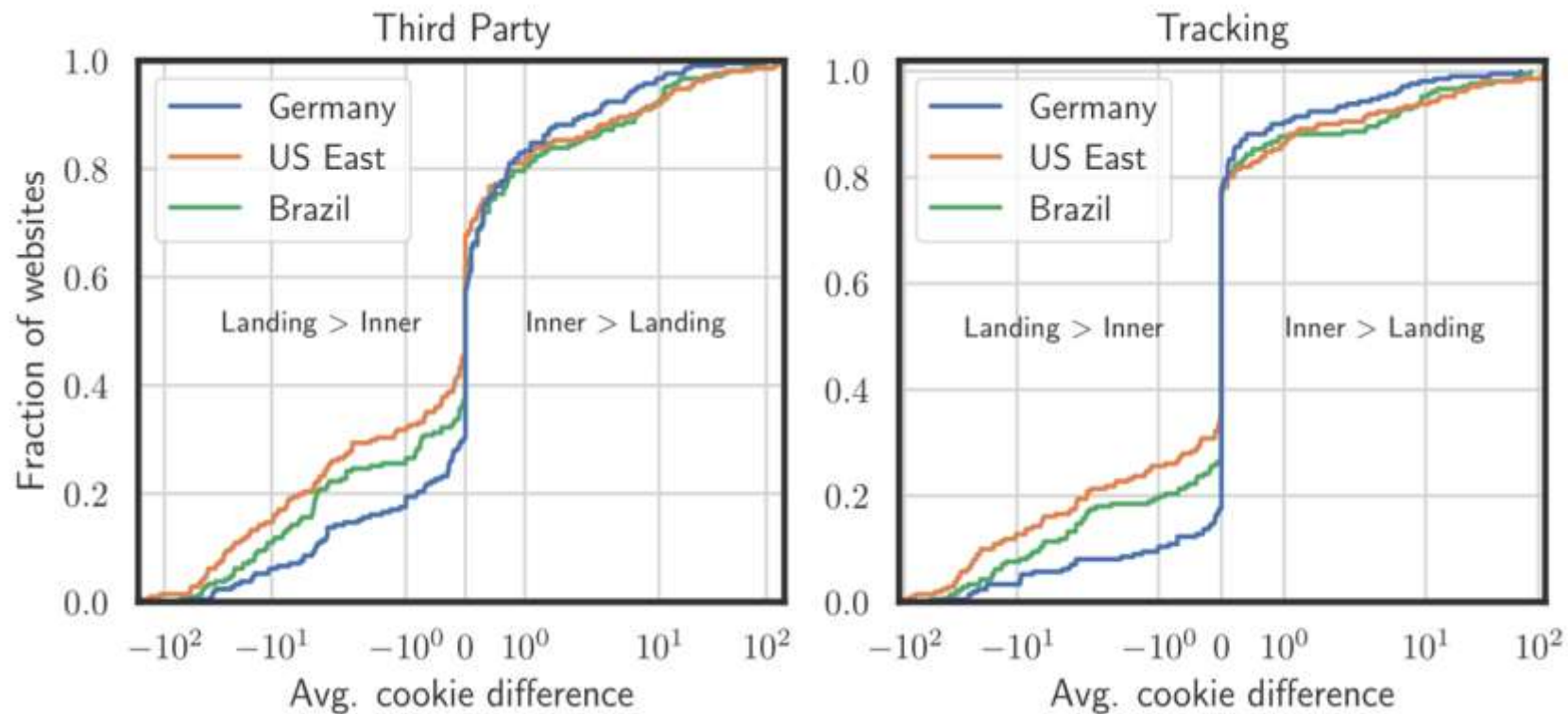


# Backup: Words Frequency



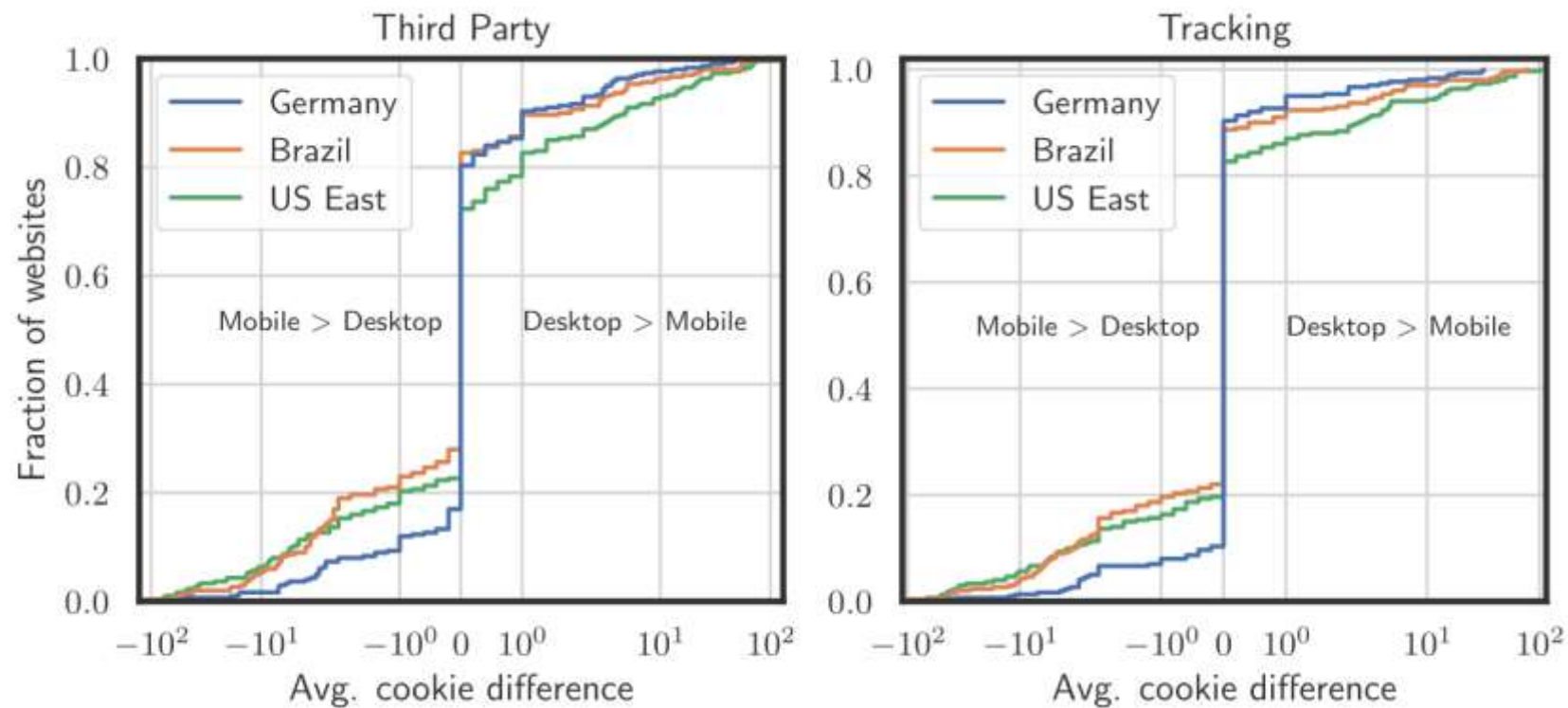


# Backup: Inner vs Landing





# Backup: Mobile vs Desktop

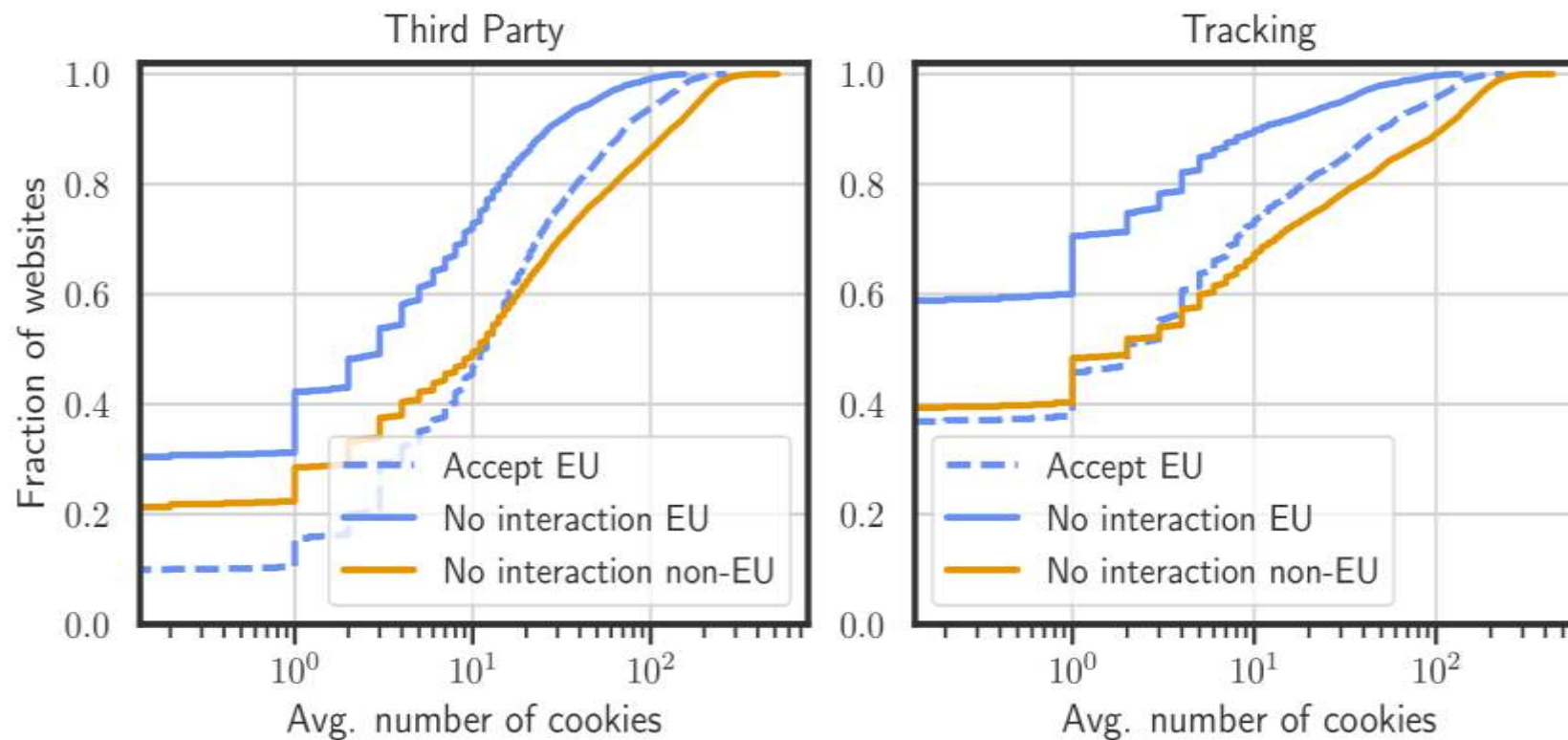


**User agent would impact the number of cookies.**





# Backup: Cookies – EU vs non-EU

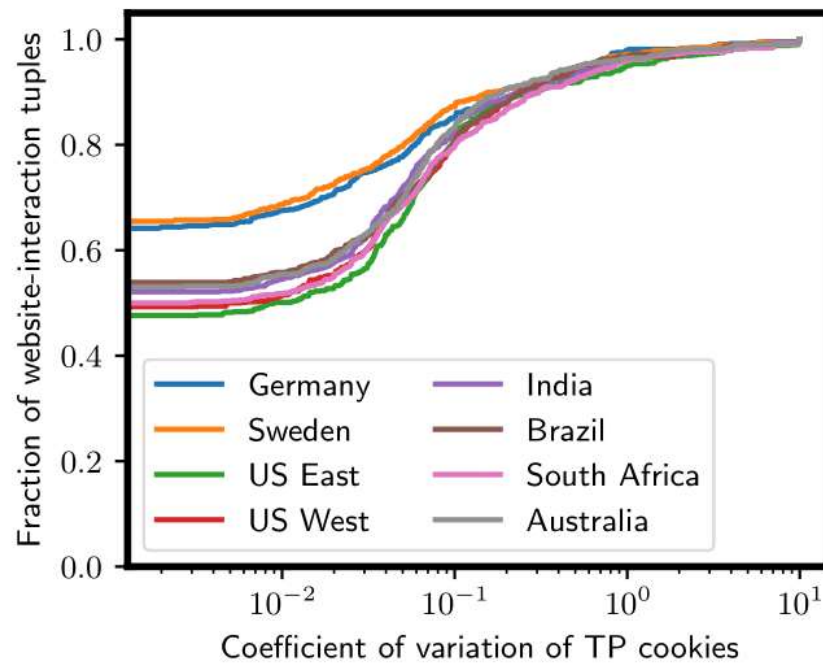


**Less cookies in EU compared to non-EU**

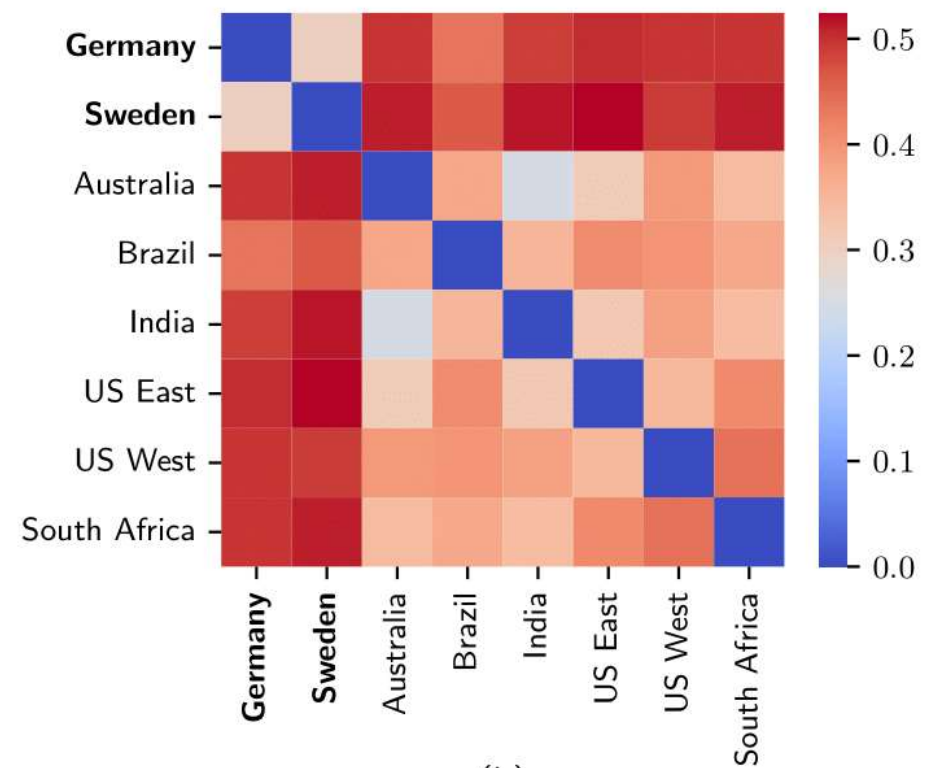




# Backup: Consistency



(a)

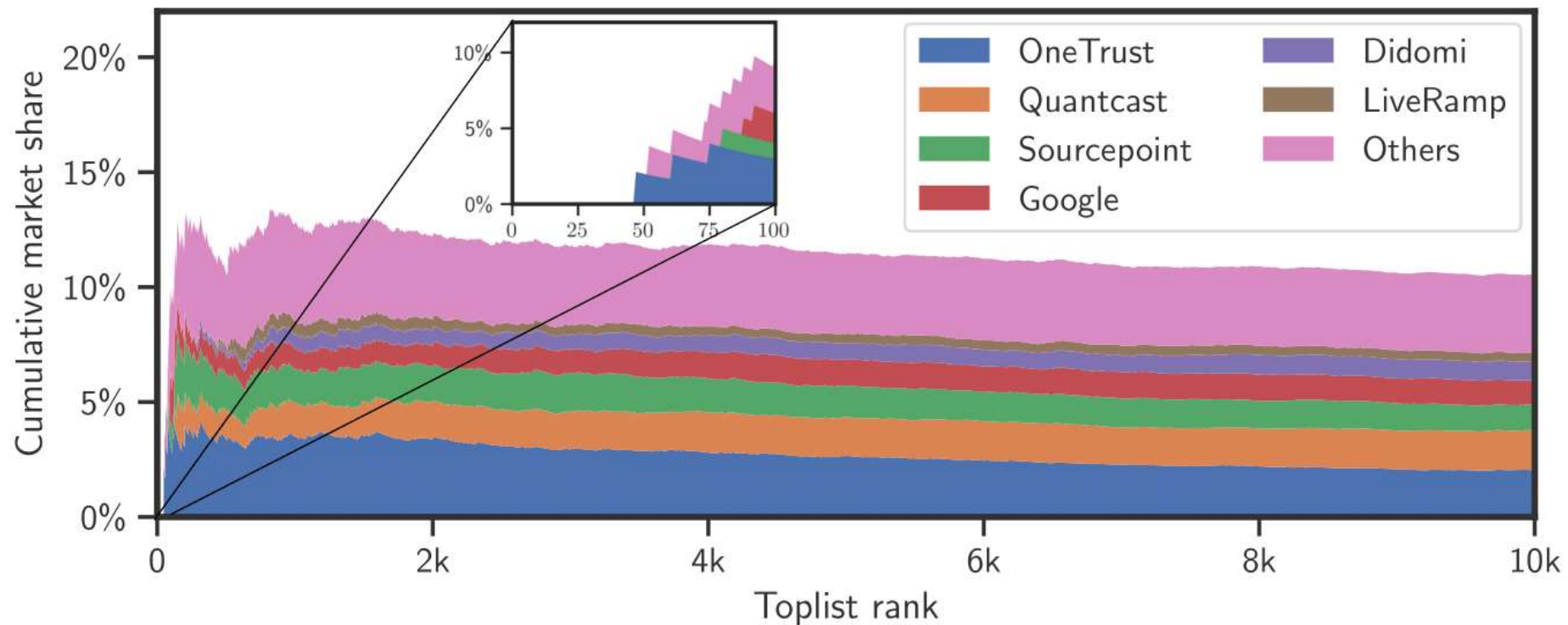


(b)



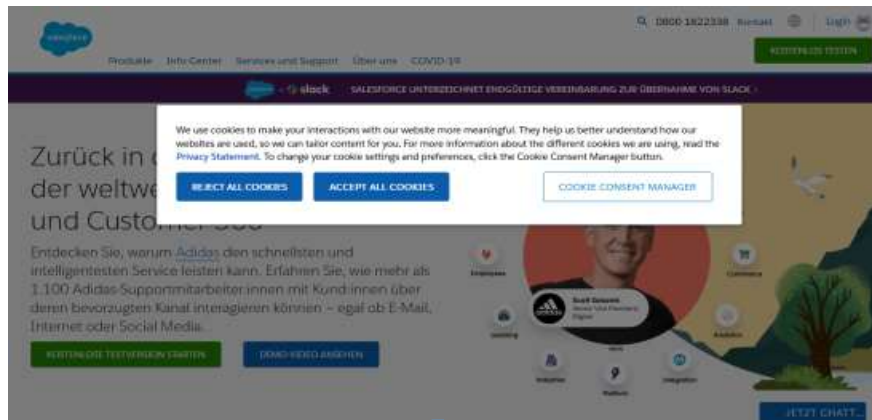


# Backup: Consent Management Platform





# Backup: Explicit vs. Implicit



We use cookies to make your interactions with our website more meaningful. They help us better understand how our websites are used, so we can tailor content for you. For more information about the different cookies we are using, read the [Privacy Statement](#). To change your cookie settings and preferences, click the Cookie Consent Manager button.

REJECT ALL COOKIES

ACCEPT ALL COOKIES

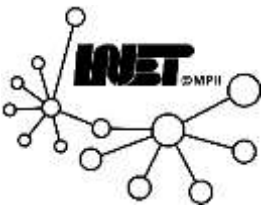
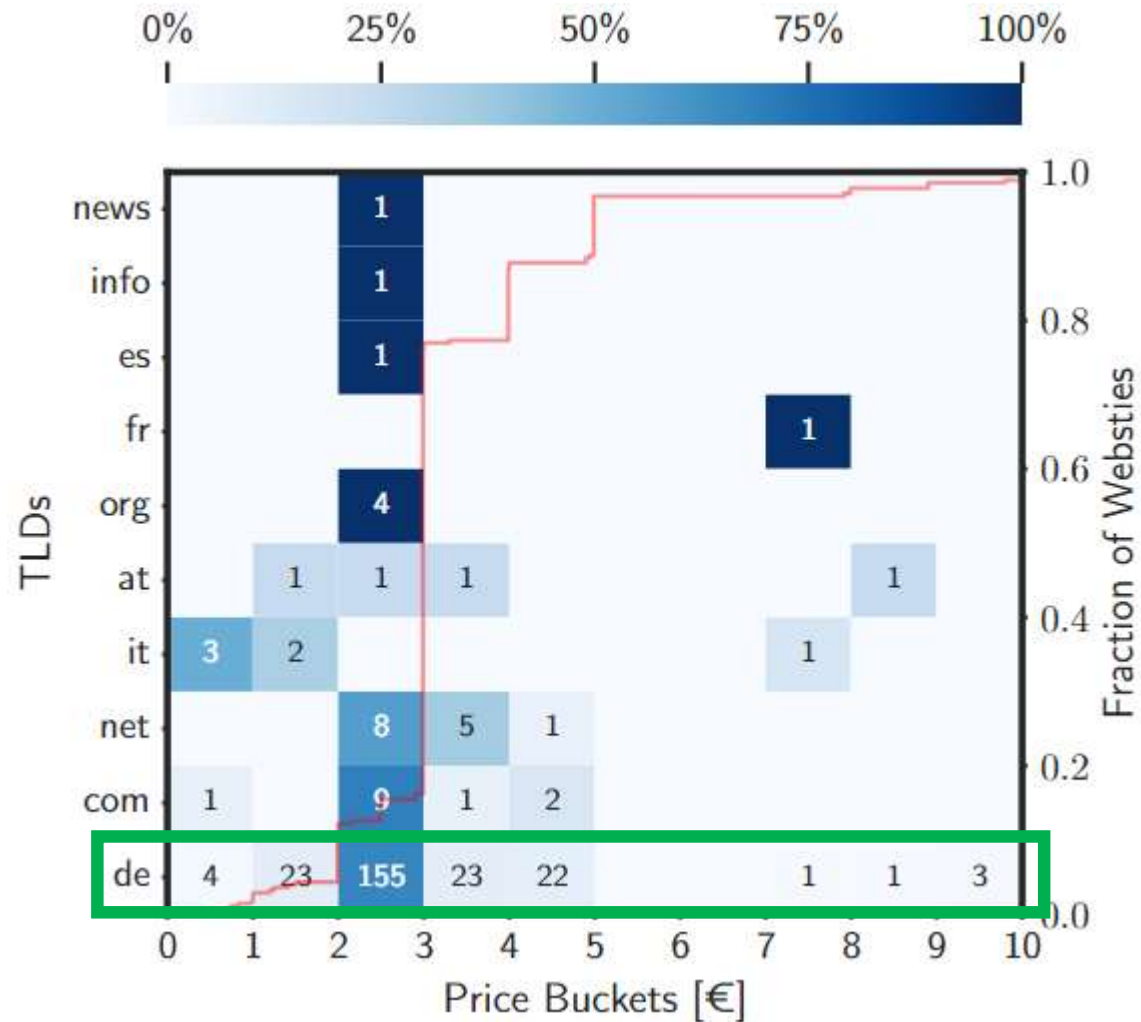
COOKIE CONSENT MANAGER



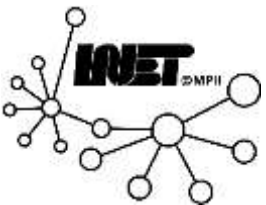
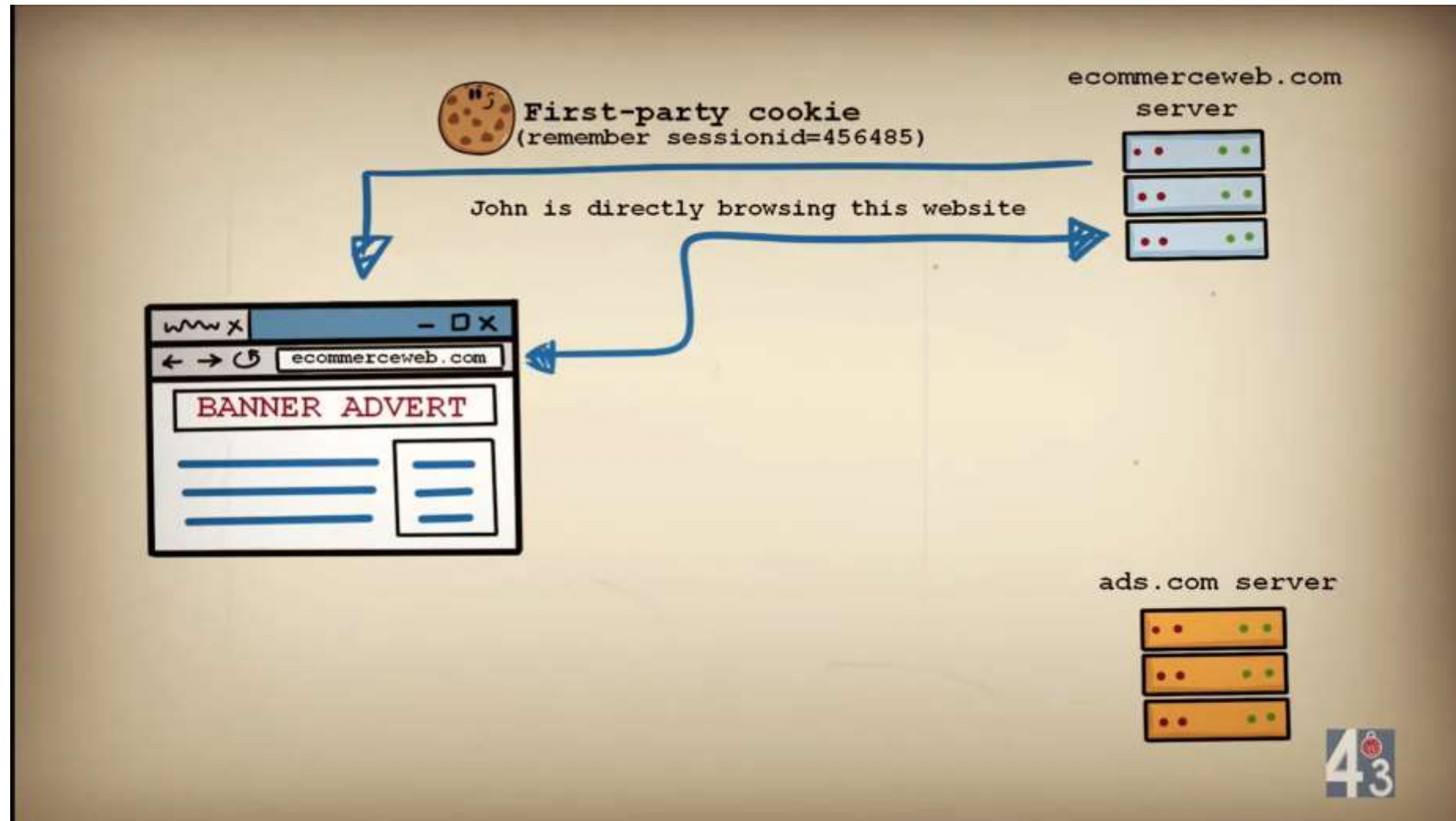
Prime Video verwendet Cookies, um Ihnen bestimmte Dienste zur Verfügung zu stellen, einschließlich Authentifizierung, Speicherung von Einstellungen und Bereitstellung von Inhalten. Andere Amazon-Webseiten können Cookies für weitere Zwecke verwenden. Weitere Informationen darüber, wie Amazon Cookies verwendet, finden Sie in den Amazon-Cookie-Hinweisen.



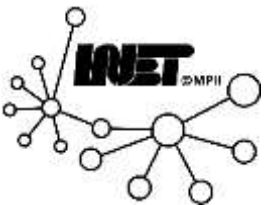
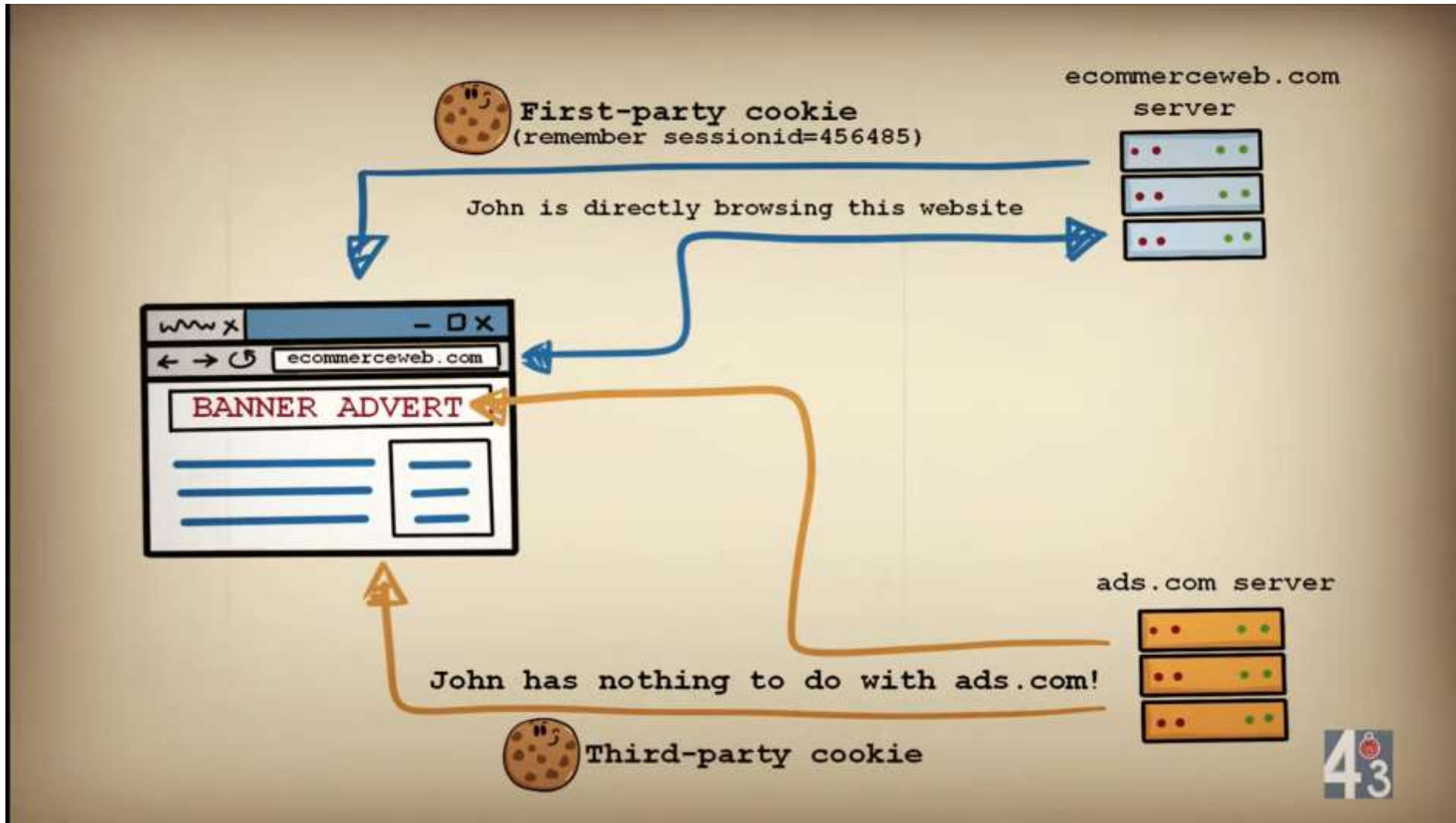
# Distribution of Monthly Subscription Price



# First-Party Cookies



# Third-Party Cookies



# Tracking Cookies

